End-to-End Data Analytics Project | Python, Power BI, Pandas

underrstanding data and attribute classification :

Dataset insurance.csv

Unit coverage understanding data and attributes classification

Tools Google colab(python . pandas. Matplotlib)

Background :

The insurance dataset contains 1,338 records describing patient demographics and their medical insurance charges. The features include age, sex, BMI, number of children, smoker status, region, and charges. This dataset provides an excellent opportunity to understand diff erent attribute types and perform foundational data processing tasks.

Tasks :

Part A – Attribute Type Identification :

Attribute	Type	Explanation
Age	Continuous	Numeric measurable value
Sex	Nominal	Category (male/female)
Bmi	continuous	Measured body index
Children	discrete	Countable values
Smoker	Nominal	Yes/No category
Region	Nominal	Categories (4 regions)
Charges	continuous	Numeric cost

PART B – Data Cleaning (WITH CODE) :

STEP 1: Open Google Colab :

https://colab.research.google.com

STEP 2: Upload Dataset :

```
from google.colab import files
uploaded = files.upload()
```

Upload file :

insurance.csv / HR_comma_sep.csv

Step 3: Load dataset : import

pandas as pd python code df =

```
pd.read_csv("insurance.csv")
df.head(10)
```

Google Colab page :

```
from google.colab import files
uploaded = files.upload()
```

Choose files insurance.csv

insurance.csv(text/csv) - 55628 bytes, last modified: 03/04/2026 - 100% done

```
Saving insurance.csv to insurance (1).csv
```

```
import pandas as pd

df = pd.read_csv("insurance.csv")
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Step 4: Check info :

df.info()

Run :

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

Note :

:- Dataset contains 1338 rows and 7 columns with no major datatype issues.

Step 5: Missing & duplicates :

df.isnull().sum()

df.duplicated().sum() Run

:

```
df.isnull().sum()
df.duplicated().sum()
```

```
np.int64(1)
```

Note :

No missing values and no duplicate records found (if true).

Step 6: Data issues :

df.describe() Run

:

```
df.describe()
```

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

Note :

No negative values observed. BMI values are within reasonable range.

Step 7: Encoding :

df['smoker'] = df['smoker'].map({'yes':1,'no':0})

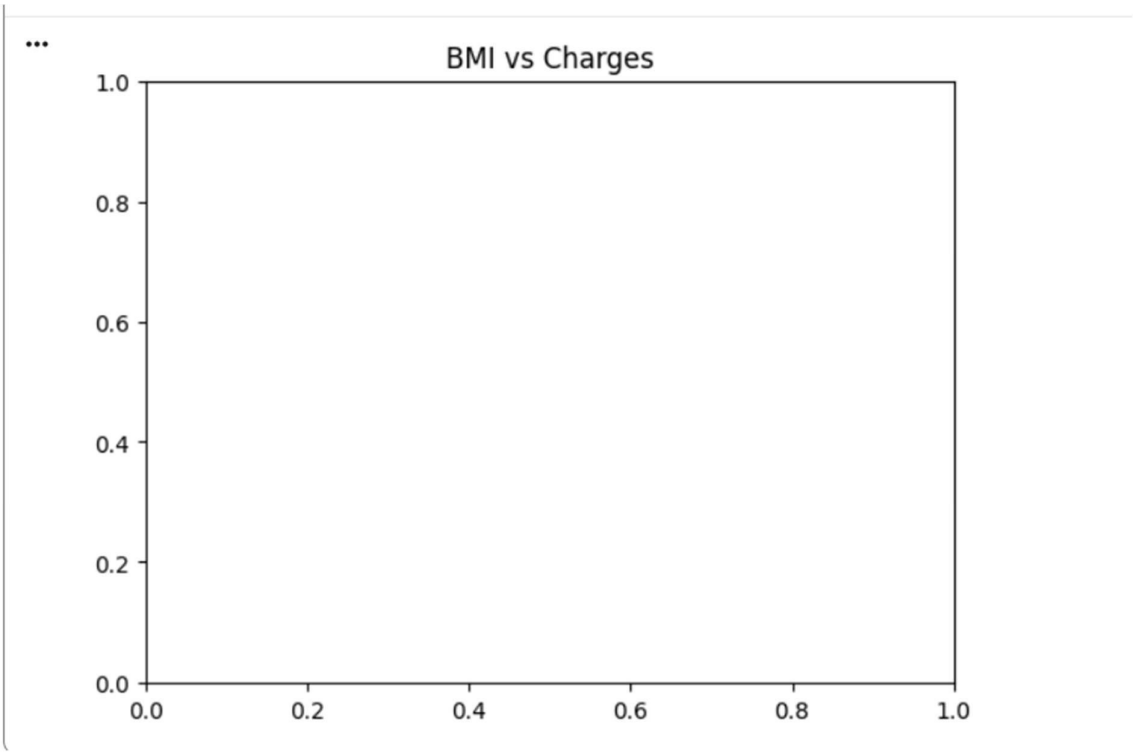
df = pd.get_dummies(df, columns=['region'])

Note :

Smoker converted to binary (0/1) and region converted using one-hot encoding.

PART C – Data Viewing :

1. Scatter Plot (BMI vs Charges): import seaborn as sns import matplotlib.pyplot as plt sns.scatterplot(x='bmi', y='charges', hue='smoker', data=df) plt.title("BMI vs Charges") plt.show() Run :



Note :
Smokers have significantly higher charges compared to non-smokers.

2. Data Matrix (First 10 rows) :

df[['age','bmi','children','charges']].head(10)

Run :

```
df[['age','bmi','children','charges']].head(10)
```

	age	bmi	children	charges
0	19	27.900	0	16884.92400
1	18	33.770	1	1725.55230
2	28	33.000	3	4449.46200
3	33	22.705	0	21984.47061
4	32	28.880	0	3866.85520
5	31	25.740	0	3756.62160
6	46	33.440	1	8240.58960
7	37	27.740	3	7281.50560
8	37	29.830	2	6406.41070
9	60	25.840	0	28923.13692

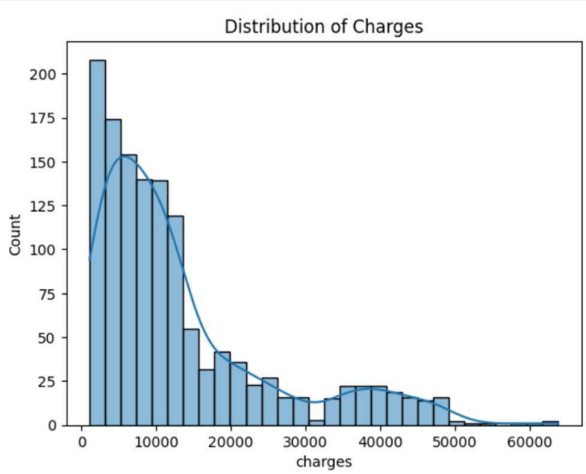
3. Histogram (Charges) :

sns.histplot(df['charges'], kde=True)

plt.title("Distribution of Charges")

plt.show() Run :

```
plt.title("Distribution of Charges")
plt.show()
```



Note :

Distribution is right-skewed (positively skewed).

PART D – Correlation Analysis :

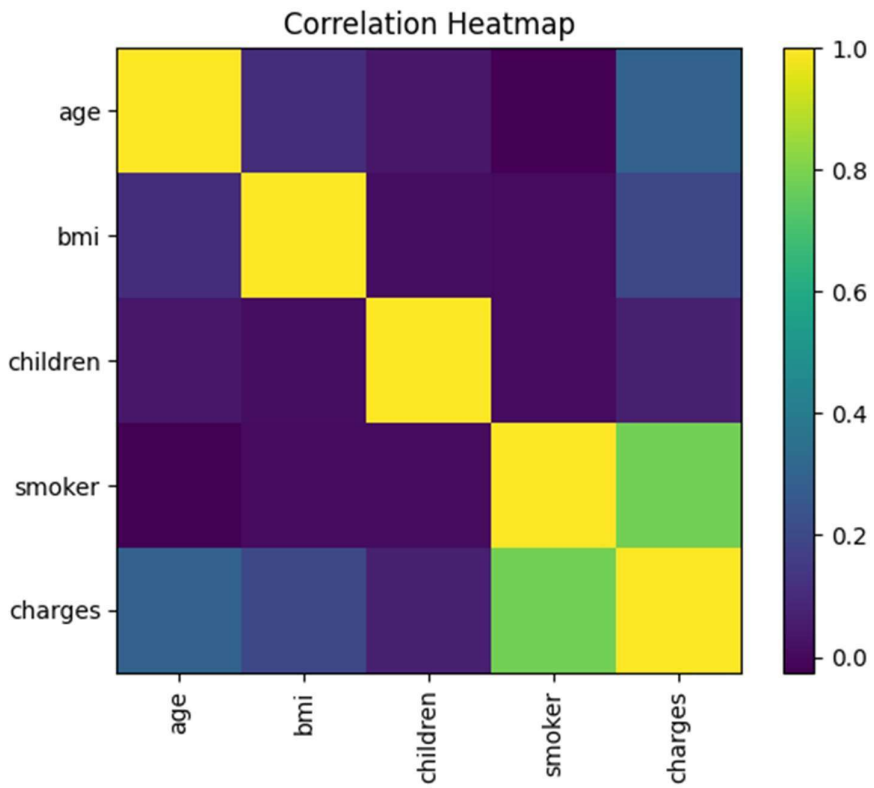
1. Correlation matrix :

corr = df.corr() corr

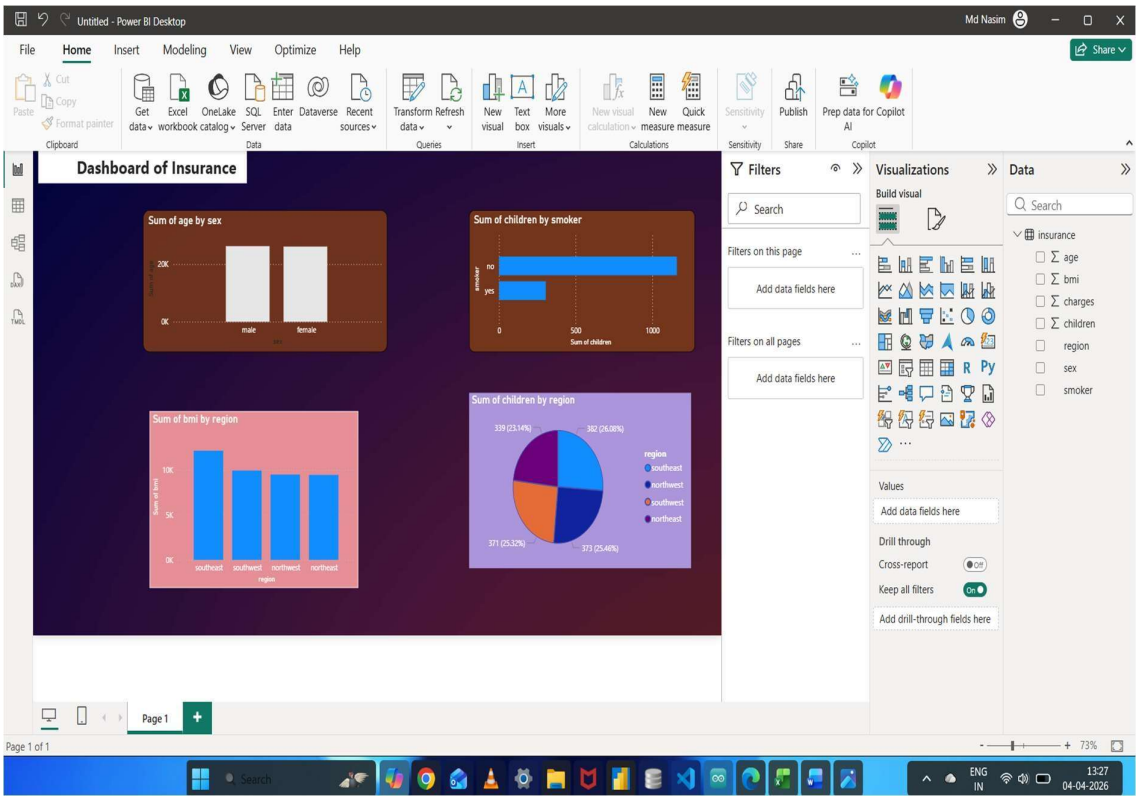
2. Heatmap :

sns.heatmap(corr, annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap") plt.show() Run :



DASHBAOARD OF INSURANCE by Using Powe bi:



Explanation about this Dashbaord :

Gender vs Age: The total age distribution between males and females is almost equal, indicating a balanced dataset with no strong gender dominance.

Smoker vs Children: Non-smokers have a significantly higher total number of children compared to smokers. This may suggest that families are more common among nonsmokers.

BMI by Region: The southeast region has the highest total BMI, while other regions (southwest, northwest, northeast) are slightly lower and relatively similar.

Children by Region: The distribution of children is fairly even across all regions, with no region showing a major spike or drop.